

SpaSE-UNet3D: Sensor-Driven Wildfire Detection and Progression Prediction from VIIRS Multispectral Imagery

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Abstract

Timely wildfire monitoring depends critically on optical and thermal infrared sensor observations from spaceborne instruments. The Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 is the primary sensor system underlying the TS-SatFire benchmark (2025), which consolidates multispectral VIIRS image stacks for three tasks: Active Fire (AF) detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. We make two contributions. First, a [systematic label-quality audit across all tasks and splits](#) reveals that many fires lack ground-truth annotations: 18 training fires and 2 test fires are excluded for AF, and the two unannotated test fires cannot be scored by any model. We further show that the reference evaluation of this benchmark differs from an independently implemented one in eight respects, and that these differences move F1 by up to 0.10 on fixed model weights—several times the spread separating the published baselines from one another. Published figures are therefore not directly comparable to independently computed values, and we make no state-of-the-art claim. We also document the BA label encoding in the released GeoTIFFs—a prerequisite for downstream modelling; [the BA task is used for auditing only and no BA model is trained](#). Second, we propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet. By using spatial-only (1, 3, 3) convolutions, our architecture avoids temporal mixing [in the convolutional path](#) on short observation windows, while utilizing SE channel attention to reweight the VIIRS spectral bands dynamically. [Under the protocol defined in this paper, SpaSE-UNet3D reaches \$F1 = \[F1 \pm SD\]\$ on fire progression over the full 24-fire test set, and \$F1 = \[F1 \pm SD\]\$ on active fire detection, in both cases from two-day observation windows where the published baselines use six. On active fire, a single-day input reaches \$F1 = \[F1 \pm SD\]\$, close enough to the two-day result that we treat the two as substantially equivalent, indicating that one VIIRS acquisition already carries most of the detectable signal—which matters operationally for early detection, before a second overpass is available. The two variants use 24.3 M and 32.9 M parameters and train in under 15 hours on a single GPU. Code and results are publicly available.](#)

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1. Introduction

Wildfires are increasingly destructive across temperate and boreal biomes, and timely automated detection of active fire fronts, burned extents, and near-term fire progression is essential for operational response. Satellite-borne optical and thermal infrared sensors

are the primary mechanisms enabling this monitoring at continental scale, with daily or twice-daily revisit and spectral bands specifically calibrated for fire radiometry. Among spaceborne systems, the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 stands out: it provides 22 bands spanning the visible, near-infrared, and thermal infrared spectrum, of which five imaging-resolution bands at 375 m are directly exploited for active fire detection [1]. Barmapoutis et al. [2] survey the full landscape of optical sensor systems for early fire detection, distinguishing terrestrial, airborne, and spaceborne architectures and confirming that VIIRS and MODIS spaceborne sensors provide the most operationally scalable coverage for large-scale wildfire monitoring. Alongside these learned approaches, index-based methods applied to moderate-resolution optical sensors remain widely used for post-fire burned-area assessment: spectral indices such as the Normalized Burn Ratio have been applied to Landsat 8 imagery to delineate burning areas in protected forest [3], and comparable index-based procedures using Sentinel-2 data have been used to quantify burned extents in wildlife sanctuaries over multi-year periods [4]. These methods provide the operational baseline against which learned approaches are measured.

The recently released TS-SatFire dataset [5] consolidates several years of VIIRS observations into a common benchmark for three wildfire tasks: Active Fire (AF) pixel detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. Each fire event is represented as a temporal stack of multispectral VIIRS sensor images, with up to eight daytime reflectance and thermal channels plus complementary nighttime thermal bands — a design motivated by the demonstrated value of combining daytime optical sensor observations with nighttime thermal infrared data for round-the-clock fire tracking [6]. The dataset paper reports results for 12 baselines ranging from 2D U-Nets [7] to 3D transformer variants [8,9], using sensor temporal stacks of up to $TS = 6$ days. The strongest published AF baseline is SwinUNETR-3D at $TS = 2$ with $F1 = 0.823$; the strongest published FP baseline is U-Net-3D at $TS = 6$ with $F1 = 0.375$.

The benchmark has been used to rank eleven architectures against one another, but neither the completeness of its labels nor the evaluation procedure used to produce those rankings has been examined. Both, we find, materially affect the reported ordering. Our contribution is therefore twofold, and we state the ordering explicitly: the primary contribution is documentation of the benchmark's evaluation protocol and label quality; the secondary contribution is an architecture, presented on efficiency grounds.

In this work we revisit TS-SatFire with two goals. The first is to examine the sensor data itself. A systematic audit of every fire in every split reveals a substantial number of fires with entirely missing labels across all three tasks, including two of the seventeen active-fire test fires. Fires that carry no annotation cannot be scored, so any test aggregate over the full split is determined for those fires by the convention adopted for empty ground truth rather than by model behaviour. We document the exclusion lists, the undocumented burned-area label encoding, and the effect of filtering at fire and day level, and we measure rather than assert the consequence for accuracy.

The second goal is architectural. The TS-SatFire sensor input is very short: operational constraints and paper conventions favour temporal windows of 1–2 days for AF and 2–6 days for FP. On such short windows, the implicit temporal prior that motivates full 3D convolutions in video understanding [10,11] or volumetric medical imaging [12] is *not obviously justified*: there is little inter-day sensor signal to decorrelate. Prior work in action recognition has shown that factorising 3D convolutions into separate spatial and temporal components yields better accuracy with fewer parameters [10,11], and *our design tests this intuition at its limit* by removing temporal mixing inside the residual blocks entirely. We propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet that explicitly uses

(1, 3, 3) convolutions in every residual block, so that the temporal dimension is preserved but never mixed across days. The block further uses SE [13] channel attention to reweight the VIIRS multispectral sensor bands per spatial context. Task-specific variants add an ASPP [14,15] bottleneck and a 27-channel auxiliary-feature input for fire progression.

We evaluate SpaSE-UNet3D on both the AF and FP tasks under the evaluation protocol defined in Section 6. On AF it reaches $F1 = [F1 \pm SD]$ at $TS = 2$ on the 15 annotated test fires, and on FP $F1 = [F1 \pm SD]$ at $TS = 2$ over the full 24-fire test set, in both cases using two-day observation windows where the published baselines use six, and training in under 12 hours on a single NVIDIA T4 accelerator. We do not present these figures as improvements over the published baselines: as Section 6 establishes, the two sets of numbers are not computed under a common protocol. We treat the BA task as audit-only: the audit itself is a dataset contribution, but we do not present a learned BA model.

The objectives of this study are: (i) to quantify the completeness of the TS-SatFire labels across all three tasks and splits, and to document the burned-area encoding required to use them; (ii) to determine whether published baseline figures and independently computed figures measure the same quantity, and to isolate the contribution of each difference; (iii) to develop an architecture suited to the short observation windows the benchmark provides, and to characterise its computational cost as well as its accuracy; and (iv) to establish the run-to-run variability of both tasks, so that reported differences can be interpreted against measurement noise.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the TS-SatFire dataset. Section 4 presents our label-quality audit. Section 5 details the SpaSE-UNet3D architecture and its task-specific variants. Section 6 reports the experimental protocol and results. Section 7 discusses the findings and Section 8 concludes.

2. Related Work

Wildfire remote sensing with deep learning.

Early work on active fire detection from satellite imagery used per-pixel thresholds on thermal bands, typically VIIRS band I4 or the MODIS analogue [1,16]. Deep learning approaches in the last five years have moved to segmentation networks operating on multi-band VIIRS or Sentinel-2 stacks [17], and more recently to 3D and transformer architectures that incorporate temporal context [18]. Huot et al. [19] released the Next Day Wildfire Spread benchmark, with a related but coarser problem formulation than TS-SatFire. Zhao and Ban [5] provide the most comprehensive comparison to date, benchmarking 12 architectures spanning U-Net [7], Attention U-Net [20], UNETR [9], SwinUNETR [8], and recurrent encoders on a unified TS-SatFire benchmark. Their reported figures are the reference point for this paper, and Section 6.2 examines the evaluation procedure that produced them.

Factorised spatiotemporal convolutions.

Tran et al. [10] and Qiu et al. [11] showed that factorising 3D convolutions into separate spatial $(1, k, k)$ and temporal $(k, 1, 1)$ stages outperforms full (k, k, k) kernels in video classification, with fewer parameters. SpaSE-UNet3D extends this principle to its extreme on short ($T \leq 2$) windows by retaining only the spatial component. Our ablation (Section 6.5) supports the parameter-efficiency half of this claim; it does not resolve an accuracy difference.

Attention in U-Nets.

Squeeze-and-Excitation [13] adds a lightweight channel-wise attention module that globally pools feature maps and produces per-channel gating weights. Roy et al. [21]

extended this with concurrent spatial-channel SE for 2D semantic segmentation. Atrous Spatial Pyramid Pooling [14], later refined in DeepLabv3+ [15], aggregates multi-scale context via parallel dilated convolutions in the bottleneck. Both modules are standard in semantic segmentation, but their combination with spatial-only 3D convolutions for temporal-satellite tasks is, to our knowledge, new.

Data quality in remote-sensing benchmarks.

Prior work has documented label noise in land-cover benchmarks [22] and machine-learning test sets more broadly [23], showing that erroneous test labels can reorder benchmark rankings even when the underlying models are unchanged. Moreno-Ruiz et al. [24] document omission and commission errors in the MODIS burned-area products from which TS-SatFire’s burned-area labels are partly derived, so label error in this benchmark is inherited as well as introduced. Our audit is in this tradition, and extends it in one respect: we find that the evaluation procedure, not only the labels, determines the reported ordering.

Sensor systems for wildfire monitoring in *Sensors*.

A body of work published in the MDPI *Sensors* journal is directly relevant to our setting. Barmpoutis et al. [2] provide a comprehensive survey of optical remote sensing systems—from ground-based infrared cameras to spaceborne VIIRS and MODIS sensors—for early fire detection, establishing the sensor taxonomy within which TS-SatFire sits. Chiang and Ulloa [6] demonstrate that combining the Sentinel-3 SLSTR optical/thermal instruments with the VIIRS day-night band (VIIRS-DNB) enables round-the-clock fire tracking, motivating TS-SatFire’s inclusion of both daytime and nighttime VIIRS channels as model inputs. For burned area assessment, Moreno-Ruiz et al. [24] audit the omission and commission errors of MODIS-derived BA products, confirming that these remotely sensed labels carry systematic spatial and temporal biases that propagate to downstream benchmarks—a finding our own audit of TS-SatFire BA labels corroborates. On the deep learning side, Ghali et al. [25] benchmark CNNs and vision transformers for pixel-level wildfire segmentation from satellite imagery, and Casallas et al. [26] apply a 3D-CNN directly to satellite stacks for wildfire early alert, providing the closest architectural precedent to our SpaSE-UNet3D in the *Sensors* literature.

3. The TS-SatFire Dataset

TS-SatFire [5] consists of a set of fire events, each represented by a multi-day temporal stack of VIIRS observations. Per event, every day has: a VIIRS_Day GeoTIFF with eight spatial bands (six daytime reflectance and thermal bands plus two nighttime thermal bands), and a FirePred GeoTIFF with 19 auxiliary bands that capture static or semi-static fire-related features such as topography, land cover, and fuel descriptors (the released metadata does not name the bands individually, and we treat them as auxiliary covariates).

3.1. Labels

The VIIRS_Day GeoTIFF contains the label channels for all three tasks, encoded in dedicated bands of the same raster.

Active Fire labels.

Band 7 contains active-fire labels, encoded as VIIRS fire confidence levels [1]. Pixels with finite values ≥ 7 are active-fire pixels, which we threshold to a binary mask.

Burned Area labels.

Band 8 contains burned-area labels. Zhao et al. [5] describe how the BA labels are constructed — as the union of accumulated VIIRS active-fire detections and NIFC perimeters, with case-by-case manual quality control on the test set. The dataset paper does not document the precise storage convention used in the released GeoTIFFs, which we found necessary to verify before any modelling: in practice, band 8 encodes the burned mask implicitly, with finite values in the range [2,119, 908,660] corresponding to Julian-day codes or perimeter identifiers and NaN values corresponding to unburned pixels. The correct binary extraction is therefore

$$BA(p) = \mathbf{1}[\neg \text{NaN}(\text{band8}(p))] \quad (1)$$

and not a value threshold. Any future BA modelling on TS-SatFire must use this encoding.

Fire Progression labels.

Fire progression labels are not stored directly. They are derived from consecutive BA masks: the FP label for day T is the set of pixels burned on day $T + 1$ but not on day T ,

$$FP(T) = BA(T + 1) \setminus BA(T). \quad (2)$$

A window is usable only when both the last input day and the following day have valid (non-NaN) band 8.

3.2. Splits

The dataset paper defines the following splits.

AF / BA splits.

138 training fires, 13 validation fires, and 17 test fires (named historical events, e.g., *Dixie*, *Creek*, *Carr*). Two of the 17 test fires, *Calf Canyon* and *Mosquito*, are 2022 events.

FP splits.

The FP validation set extends the AF/BA list from 13 to 15 fires, adding 22713339 and 23301962; since 22713339 is present in the dataset, FP training has 137 fires rather than 138. 23301962 is absent from the Kaggle distribution, so the replicable FP validation set contains 14 fires. The test set is a separate collection of 24 US_2021 fires identified by geographic codes such as US_2021_CA3451712013120211011.

4. Label-Quality Audit

We audit every fire in every split of every task for label completeness. Missing labels occur at two distinct granularities, and conflating them discards usable data: some fires carry no annotation on any day of their temporal extent, while others are annotated on a subset of days only. We therefore audit and filter at both levels. For each fire we read the VIIRS_Day GeoTIFF for every available day and test whether the relevant label band contains any finite pixel. A fire with zero finite label pixels across its entire temporal extent carries no supervision and is excluded; a fire annotated on some days is retained, and individual unlabelled windows are rejected instead.

4.1. AF Audit

Table 1 summarises the AF audit.

Fire level. Of the 138 fires in the AF training pool, 18 have band 7 entirely NaN across every available day and are excluded; one validation fire (22712904) and two test fires fail

the same test. The two test fires are *Calf Canyon* and *Mosquito*, both 2022 events for which active-fire ground truth was never included in the released data. They cannot be scored by any model, and any aggregate reported over the full 17-fire test set therefore depends for those fires on the convention adopted for empty ground truth rather than on model behaviour (Section 6.2). The audited splits contain 120 training, 12 validation and 15 test fires.

Day level. Excluding a fire outright because part of its temporal extent is unlabelled discards usable supervision. We instead retain partially annotated fires and reject individual windows: a window is admitted only if band 7 on its final day—the day to which the AF label refers—contains at least one finite pixel. Applied to the audited fire lists this rejects 143 training windows, leaving 1,719 training windows at TS = 2, each supervised by a verified label.

Without the fire-level exclusion, windows drawn from unannotated fires enter training with an all-negative target, since a NaN band read through the reference loader’s `nan_to_num` conversion becomes a valid all-zero mask. The model is then supervised to suppress activations on scenes that do contain fire. Section 6.3.1 measures the size of this effect under our pipeline.

Sensitivity to the criterion. The fire-level criterion above is the weakest defensible one: a fire is removed only when it carries no supervision at all. A stricter alternative removes any fire whose labelled days fall below a fixed fraction of its temporal extent; at a 50% cut-off, three further fires are removed, leaving 117. We adopt the weaker criterion because day-level filtering already guarantees that every training window is verified, making fire-level removal of partially labelled fires unnecessary and wasteful of supervision.

4.2. BA Audit

Table 2 summarises the BA audit. 49 of 138 training fires are excluded from BA training: 14 have no VIIRS_Day directory, 18 have band 8 entirely NaN across all days, and 17 have fewer than 50% of days with usable labels. All 5 excluded validation fires have band 8 entirely NaN. The AF and BA exclusion lists overlap substantially: 19 fires are excluded from both tasks, while 3 AF-excluded fires retain valid BA labels and 49 BA-excluded fires retain valid AF labels.

In addition to the missing-label audit, we also document the practical encoding of band 8 in the released GeoTIFFs. Zhao et al. [5] describe how the labels are produced (union of accumulated VIIRS active-fire detections and NIFC perimeters, with manual quality control on the test set) but do not specify the storage convention in the released rasters — a small but important detail for downstream users. Empirically we find that finite values in band 8 span the range [2,119, 908,660], consistent with Julian-day codes or perimeter identifiers from the MCD64A1 MODIS burned-area product rather than a binary or thresholded indicator. 100% of finite pixels are burned pixels; there are no finite non-burn values. The correct extraction of a binary BA mask is therefore presence-of-finite, as stated in Section 3. Documenting this encoding is a prerequisite for any future BA modelling on TS-SatFire.

4.3. FP Audit

Table 3 summarises the FP audit. The FP audit is more subtle because FP labels are derived at the window level from consecutive BA masks. A fire is excluded if no usable window can be constructed, i.e., if it has zero consecutive (day T , day $T + 1$) pairs in which both days have valid BA. 48 training fires and 6 validation fires fail this test. The submitted version of this paper additionally excluded eight fires from the test set, on a stated criterion of burned-area fraction below 0.001% inside the 256×256 crop. That criterion does not

describe what was done: three of the eight exceed it, and the boundary is not monotone in the quantity it names. We therefore remove the test-set exclusion entirely and report all 24 test fires throughout. Six of the excluded fires do carry no usable window and contribute no counts under micro-averaging; the remaining two are scored normally.

All exclusion lists are reported in full in the supplementary audit reports released with the repository.

Table 1. Active Fire label audit. Fire-level exclusions only; a further 143 training windows are rejected at day level.

Split	Total	Usable	Excluded
Train	138	120	18
Val	13	12	1
Test	17	15	2 (Calf Canyon, Mosquito)

Table 2. Burned Area label audit.

Split	Total	Usable	Excluded
Train	138	89	49
Val	13	8	5
Test	24 US_2021	18	6

Table 3. Fire Progression label audit. All 24 test fires are reported in Section 6; the six without a usable window contribute no counts under micro-averaging.

Split	Total	Usable	Excluded
Train	137	89	48
Val	14 (Kaggle)	8	6 (no usable window)
Test	24 US_2021	18	6 (no usable window)

5. Method: SpaSE-UNet3D

SpaSE-UNet3D is a family of two closely related encoder–decoder architectures, one for AF detection and one for FP prediction. Both variants share four design principles: (i) spatial-only (1,3,3) convolutions, (ii) SE channel attention [13] inside every residual block, (iii) a symmetric encoder/decoder with skip concatenations as in U-Net [7,12], and (iv) a combined Dice + Focal loss [27,28]. Task-specific variations handle the different input dimensionality, label density, and temporal-context budget of AF and FP. Fig. 1 shows the shared residual block side-by-side; Figs. 2 and 3 show the AF and FP variants respectively.

5.1. Shared Building Blocks

Spatial-only convolutions.

Every convolution in the encoder and decoder stages has kernel shape (1,3,3) with padding (0,1,1). The temporal dimension T is therefore preserved through all encoder/decoder stages and never mixed. For the short temporal windows used here ($T \in \{1,2\}$ for AF, $T = 2$ for FP), temporal mixing adds parameters without providing signal; our internal ablations confirmed that full (3,3,3) kernels hurt validation F1 on TS = 2.

SE channel attention.

After the second convolution in every residual block, features pass through a SEBlock3D module with reduction ratio $r = 8$. The module applies global average pooling over (T, H, W) , a two-layer MLP with dimensions $(ch, ch/r, ch)$, and a sigmoid to produce per-channel gating weights which multiply the input. This lets the network learn which VIIRS spectral bands or auxiliary features are informative per spatial context.

Downsampling and upsampling.

Encoder stages are separated by MaxPool3D(1,2,2), which halves the spatial dimensions but leaves T unchanged. Decoder stages upsample with ConvTranspose3D(1,2,2), and the upsampled features are concatenated with the matching-resolution encoder skip.

Residual blocks.

Each block applies two $(1,3,3)$ convolutions with batch normalisation [29] and a non-linear activation, followed by SE attention and a residual addition with the (possibly projected) input. Both variants of the ResBlock3D are shown in Fig. 1. The two variants differ only in the choice of activation (ReLU vs. GELU [30]) and the presence or absence of Dropout3D [31].

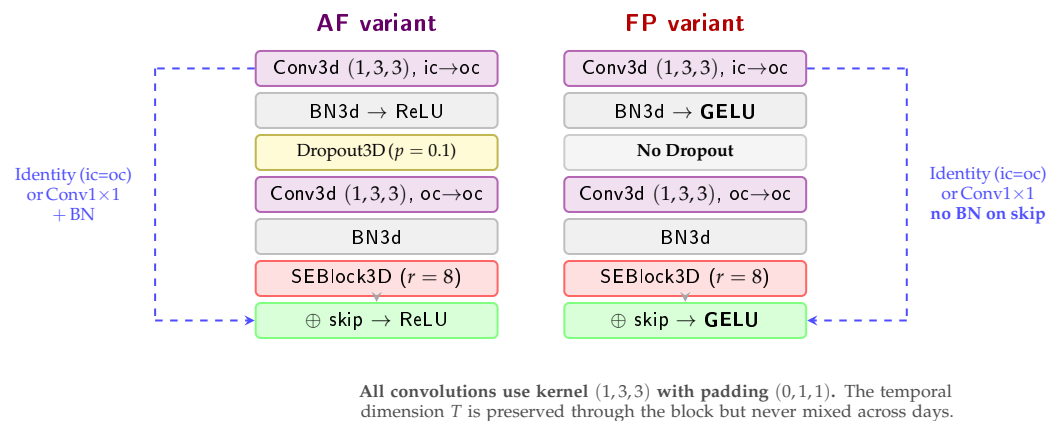


Figure 1. Internal structure of the ResBlock3D used in both SpaSE-UNet3D variants. Both blocks share spatial-only $(1,3,3)$ convolutions and SE channel attention. The AF variant (left) uses ReLU activations and Dropout3D for regularisation; the FP variant (right) uses GELU and removes Dropout, which empirically hurt convergence on the smaller, sparser FP training set. The skip path either passes the input unchanged (when $ic = oc$) or applies a $1 \times 1 \times 1$ projection convolution.

5.2. AF Variant

The AF variant takes an 8-channel input: the six daytime VIIRS bands and two nighttime VIIRS bands, stacked along the channel dimension of the temporal tensor. The encoder has four stages with output channels [64, 128, 256, 512]. The bottleneck is a single ResBlock3D that widens to 1024 channels. The decoder mirrors the encoder with channels [512, 256, 128, 64], each stage being a ResBlock3D operating on the concatenation of the upsampled feature map and the encoder skip. The full architecture is shown in Fig. 2.

ResBlock3D (AF).

The AF residual block (Fig. 1, left) applies two $(1,3,3)$ convolutions with batch normalisation, ReLU activation, Dropout3D ($p = 0.1$) between them, an SE channel-attention module, and a residual addition with a projected (or identity) skip [32]. The Dropout3D

inside the block regularises the network against the relatively small effective training set of 120 fires.

Deep supervision.

To force intermediate decoder representations to be spatially specific, the AF variant attaches an auxiliary segmentation head to the third decoder stage (d_3 , 256 channels), upsamples its output to the input resolution, and adds its loss to the total objective with weight $DS_WEIGHT = 0.3$. Both the main and auxiliary losses are computed on the last temporal frame only: the model produces predictions for all T input days, but the AF labels are only available for the final day, so we supervise only that frame.

Loss.

The AF training objective is a convex combination of Dice [27] and Focal loss [28]:

$$\mathcal{L}_{AF} = 0.5 \mathcal{L}_{Dice} + 0.5 \mathcal{L}_{Focal}(\alpha=0.75, \gamma=2) + 0.3 \mathcal{L}_{DS}. \quad (3)$$

The Dice term directly optimises the F1 surrogate; the Focal term handles the heavy class imbalance (fire pixels are typically $< 1\%$ of the patch).

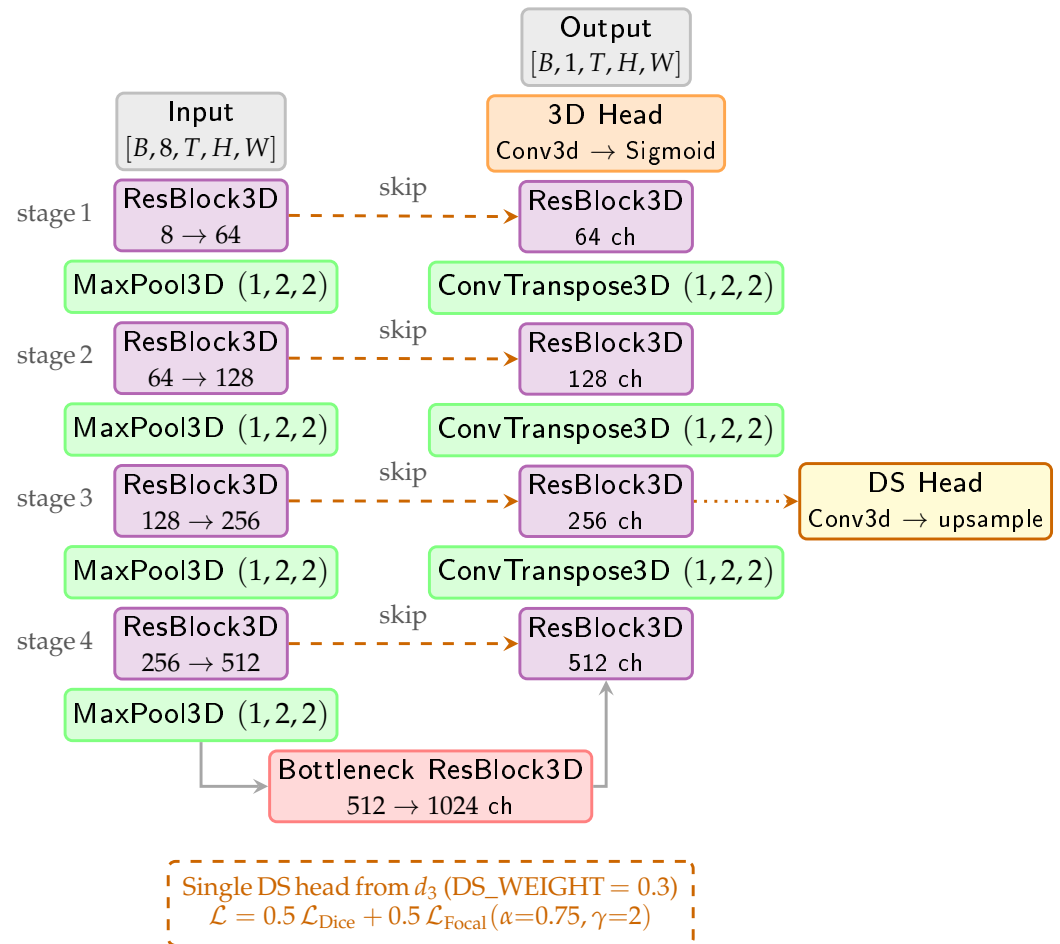


Figure 2. SpaSE-UNet3D AF variant. The encoder (left column) takes an 8-channel VIIRS input and downsamples spatially through four ResBlock3D stages with channels [64, 128, 256, 512]. A single ResBlock3D bottleneck widens to 1024 channels. The decoder (right column) mirrors the encoder with channel concatenation at every skip. A 3D head produces per-day fire masks. Deep supervision is applied at the third decoder stage.

5.3. FP Variant

The FP variant takes a 27-channel input: the 8 VIIRS bands, 18 FirePred auxiliary bands (band 3 is excluded: it is identically zero in the fire used to probe the band layout, and we retain the exclusion for consistency across runs), and 1 cumulative burned-area mask computed from all days up to T . The encoder follows the same four-stage schedule [64, 128, 256, 512], but is preceded by a dedicated input-projection layer $\text{Conv3d}(27, 64, (1, 3, 3)) \rightarrow \text{BN} \rightarrow \text{GELU}$ that mixes the heterogeneous input channels before the first encoder block. The full architecture is shown in Fig. 3.

ResBlock3D (FP).

The FP residual block (Fig. 1, right) differs from the AF block in two ways. First, GELU [30] replaces ReLU as the activation; empirically we found that the sparser loss landscape of the FP objective benefits from the smoother gradient of GELU. Second, there is no Dropout3D [31] inside the block: on the much smaller effective FP training set (89 fires, with sparse window-level positives), Dropout hurt convergence. The skip connection also drops the BatchNorm on the projection to keep the residual path closer to identity.

ASPP3D bottleneck.

The FP variant replaces the plain bottleneck of the AF variant with a 3D Atrous Spatial Pyramid Pooling module [14,15]. Four parallel branches operate on the encoder output at stride 16:

- $\text{Conv3d}(512, 341, (1, 3, 3))$ with spatial dilation rate 1;
- the same with spatial dilation rate 6;
- the same with spatial dilation rate 12;
- global average pooling over (H, W) (preserving the temporal dimension T) followed by $\text{Conv3d}(512, 341, 1)$ and broadcast back to the spatial grid.

The four branch outputs (each 341 channels) are concatenated (for $4 \times 341 = 1364$ channels) and fused with a $\text{Conv3d}(1364, 1024, 1) \rightarrow \text{BN} \rightarrow \text{GELU}$. The dilated branches capture context at multiple spatial scales, which is critical because fire-front widths in the TS-SatFire test set range from a few newly burned pixels to hundreds.

Final head.

The FP head first collapses the temporal dimension with a mean over T , then applies a 2D head $\text{Conv2d}(64, 32, 3) \rightarrow \text{BN2d} \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32, 1, 1) \rightarrow \text{Sigmoid}$. The collapse is appropriate because the FP label itself is 2D (a single-day progression mask), while the AF head remains 3D because AF labels are per-day.

No deep supervision.

We tested deep supervision for FP and removed it. The sparse, imbalanced FP labels combined with the small effective training set made auxiliary losses introduce conflicting gradients that hurt convergence.

Loss.

The FP training objective extends the AF loss with a weighted binary-cross-entropy term:

$$\begin{aligned} \mathcal{L}_{\text{FP}} = & 0.5 \mathcal{L}_{\text{Dice}} + 0.3 \mathcal{L}_{\text{Focal}}(\alpha=0.85, \gamma=3) \\ & + 0.2 \mathcal{L}_{\text{BCE}}(w_+). \end{aligned} \quad (4)$$

The addition of BCE is necessary because the FP target is far sparser than AF: typical windows have $< 0.05\%$ positive pixels. The positive weight w_+ is not a hand-tuned constant: it is computed at preload time from the observed positive-pixel ratio of the training split, and therefore varies slightly between runs (116.6–117.3 across our runs). This removes one hyperparameter from the design and ties the term directly to the class balance it is intended to correct. The higher Focal $\gamma = 3$ (vs. $\gamma = 2$ in AF) further downweights easy negatives.

Test-time augmentation.

At inference, the FP variant uses $8\times$ test-time augmentation [33]: the input is passed through horizontal flip, vertical flip, both, and three 90° rotations, plus the identity, and the resulting logits are averaged before the sigmoid and threshold.

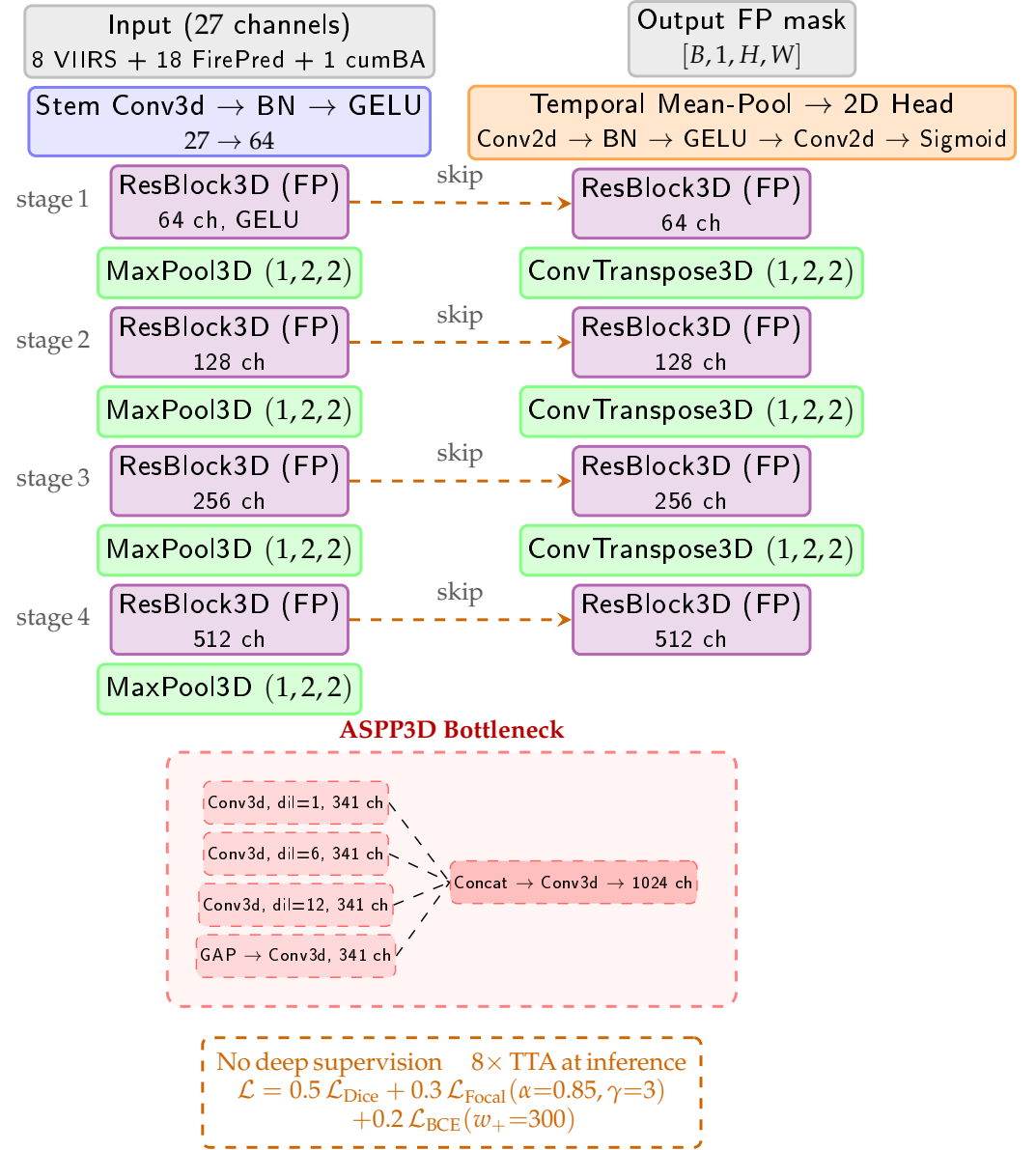


Figure 3. SpaSE-UNet3D FP variant. The encoder takes a 27-channel input combining VIIRS, FirePred auxiliary bands, and a cumulative burned-area mask. The bottleneck is replaced with an ASPP3D module that aggregates multi-scale context via parallel dilated convolutions and a global average pool. The final head collapses the temporal dimension via a mean-pool and applies a 2D convolutional head to produce a single-day FP mask. No deep supervision is used; 8× test-time augmentation is applied at inference.

6. Experiments

6.1. Implementation Details

All experiments are conducted on a single NVIDIA T4 GPU (16 GB VRAM) using PyTorch 2.10 with automatic mixed precision (AMP). A single training run takes between 3.5 and 6.5 hours depending on task and window length (Table 9); one run of each of the three configurations completes in approximately 14 hours.

Shared hyperparameters.

Batch size 8; patch size 256×256 (centre crop of the full scene); optimiser AdamW [34, 35] with learning rate 5×10^{-4} and weight decay 1×10^{-4} ; OneCycleLR [36] schedule with pct_start = 0.1 and cosine annealing; gradient clipping at norm 1.0. Each configuration

is trained from [N] (AF) and [N] (FP) independent random seeds; all reported figures are means over those runs with the standard deviation given alongside, and per-seed outputs are released with the code.

AF.

80 epochs for TS = 2, 100 epochs for TS = 1. VIIRS channels are normalised with per-band means and standard deviations computed once on a probe fire and reused across runs.

FP.

60 epochs. VIIRS channels use the same AF normalisation; FirePred auxiliary bands are normalised with per-band statistics computed at preload time on the training split and cached so that validation and test evaluation use identical statistics. Training data is augmented with horizontal and vertical flips plus 90° rotations; test-time augmentation is applied at inference as described in Section 5.

Sampling.

For AF we keep only training windows that contain at least 10 fire pixels and include at most 2 times as many negative windows as positive. For FP we use at least 3 burned pixels as the positive threshold and a negative-to-positive ratio of 1.5. These heuristics stabilise training without overfitting to the positive class.

Threshold selection.

For both tasks we sweep the inference threshold over the validation set and select the value that maximises validation F1, independently in each run, then apply it unmodified at test time. No test-set information enters the selection. The selected AF value varies widely between runs while test F1 does not, which we report as evidence of insensitivity in Section 6.3; the FP runs selected the same value throughout.

Metrics.

For all tasks we report pixel-level F1 and IoU (Jaccard index) aggregated over all pixels in all test fires (micro-average). Precision and recall at the selected operating point, and per-fire F1 and IoU, are also reported. Section 6.2 states the full evaluation protocol and its relationship to the reference implementation.

6.2. Evaluation Protocol

All baseline figures in this paper are quoted from Zhao et al. [5]; we did not retrain or re-evaluate them. Since our evaluation was implemented independently, we first ask whether the two sets of numbers measure the same quantity. They do not.

The clearest evidence needs no code. For metrics accumulated over a pixel population, F1 and the Jaccard index are related by a deterministic identity,

$$\text{IoU} = \frac{\text{F1}}{2 - \text{F1}}, \quad (5)$$

which our own measurements satisfy to four decimal places on both tasks. Every published TS-SatFire pair violates it, always in the same direction: SwinUNETR-3D at TS = 2 reports F1 = 0.823, implying IoU = 0.699 against a reported 0.727; U-Net-3D at TS = 6 reports F1 = 0.375, implying 0.231 against a reported 0.338. A one-sided violation of this form is the signature of averaging a convex function of per-unit counts rather than aggregating counts first. This is a property of the aggregation, not a transcription error, and it means the published pairs and ours are different quantities.

Reading the reference implementation confirms the mechanism and reveals seven further differences (Table 4). Two are substantive beyond aggregation. The `zero_division` convention awards a perfect score to any frame with empty ground truth—including the two test fires that carry no active-fire annotation at all (Section 4), so a model is credited for scenes on which it cannot be evaluated. And the AF label rule differs in what counts as fire: `nan_to_num(band 7) > 0` admits any non-zero confidence value, whereas the VIIRS documentation [1] places the low-confidence boundary at 7.

Table 4. Differences between the TS-SatFire reference evaluation and the protocol used in this work.

Aspect	Reference implementation	This work
Aggregation	per-frame F1, averaged over frames, then fires	counts accumulated over all pixels, metric computed once
Empty ground truth	scored 1.0 (<code>zero_division=1.0</code>)	contributes no counts
AF label rule	<code>nan_to_num(band 7) > 0</code>	$\text{band } 7 \geq 7$
FP target	accumulated active fire, or its union with burned area, selected per fire by <code>label_sel</code>	$\text{clip}(\text{BA}(T+1) - \text{BA}(T), 0, 1)$
Spatial crop	30% offset	centred
NaN in band 8	counted as burned	treated as unburned
Threshold	fixed at 0.5	selected on validation
Test stride	stride = TS; all T frames scored	stride 1, final frame scored

Because these differences are separable, their cost can be measured. Table 5 applies them cumulatively to our AF model with the weights held fixed, so that only the measurement changes.

Table 5. Our AF model evaluated under progressively more of the reference protocol; model weights fixed throughout.

Configuration	Test fires	F1
This work (final frame, $\text{band } 7 \geq 7$)	15	0.7917
+ all frames scored	15	0.7714
+ reference label rule ($\text{band } 7 > 0$)	15	0.7567
+ the two unannotated test fires	17	0.6912
<i>SwinUNETR-3D, $TS = 2$, as published</i>	17	0.823

Protocol choices thus move F1 by 0.10 on fixed weights—several times the spread separating the twelve published baselines from one another (0.713–0.823). Rankings on this benchmark are therefore not robust to evaluation convention, and our own figures are not directly comparable to the published values. We report them under the protocol stated below and do not present them as improvements.

We evaluated our model under the reference protocol rather than retraining the baselines under ours because a reimplementing would introduce our own training choices as confounds, whereas holding fixed weights and varying only the measurement isolates the protocol effect. A full retraining study under a common protocol remains valuable future work.

All results below therefore use: counts accumulated over all pixels with the metric computed once; unannotated fires excluded and reported separately rather than scored; AF labels as band 7 ≥ 7 on the final day of each window; FP targets as the clipped increment between consecutive burned-area masks; a centred 256×256 crop; and a threshold selected on validation and applied unmodified at test time. Per-seed outputs are released with the code.

6.3. Active Fire Results

Table 6 reports our test-set F1 and IoU at TS = 1 and TS = 2 alongside the twelve baselines of the dataset paper [5]. The two groups are separated because they are not computed under a common protocol (Section 6.2): the baseline figures are reproduced as published, under the reference evaluation, while ours are micro-averaged over the 15 annotated test fires at a validation-selected threshold. The table is therefore a summary of the published landscape next to an independently measured result, not a ranking, and we draw no comparative conclusion from it.

Under our protocol, SpaSE-UNet3D reaches F1 = $[F1 \pm SD]$ at TS = 2 over $[N]$ independent runs, and F1 = $[F1 \pm SD]$ at TS = 1. The difference between the two is small, and of the same order as the spread between repeated runs of the same configuration, so we treat the one-day and two-day results as substantially equivalent rather than ordering them. Per-fire inspection shows this equivalence is not uniform: individual fires shift by up to 0.022 in either direction between TS = 1 and TS = 2, with the largest losses on the weakest-signal events (Swedish, Blue Ridge). The second day therefore appears to add and remove information in roughly equal measure, which is consistent with there being little fire-specific signal in the additional frame.

Table 6. Active Fire detection results. Baseline figures are reproduced from [5] under the reference evaluation protocol; our figures are measured under the protocol of Section 6.2 on the 15 annotated test fires. The two groups are not directly comparable.

Model	TS	F1	IoU
<i>Published baselines, reference protocol, 17 test fires</i>			
U-Net (2D)	1	0.731	0.605
Att-U-Net (2D)	1	0.763	0.648
UNETR-2D	1	0.733	0.621
SwinUNETR-2D	1	0.774	0.660
GRU-3	6	0.713	0.601
LSTM-3	6	0.765	0.654
T4Fire	6	0.802	0.700
U-Net-3D	6	0.748	0.628
Att-U-Net-3D	6	0.770	0.654
UNETR-3D	6	0.811	0.706
SwinUNETR-3D	6	0.797	0.688
SwinUNETR-3D	2	0.823	0.727
<i>This work, protocol of Section 6.2, 15 annotated test fires</i>			
SpaSE-UNet3D	1	$[F1 \pm SD]$	$[IoU \pm SD]$
SpaSE-UNet3D	2	$[F1 \pm SD]$	$[IoU \pm SD]$

Threshold selection is performed independently in each run by sweeping the validation split and retaining the value that maximises validation F1, which is then applied unmodified at test time. The selected values differ substantially between runs while test F1

does not, which is itself the relevant result: performance is essentially flat in the threshold. A sweep over the test set confirms this, with F1 varying by less than 0.005 across the entire range [0.20, 0.80] as precision and recall trade off symmetrically. In every run the validation-selected threshold yielded test F1 within 0.0003 of the best achievable on the test set, so no advantage accrues from the selection procedure; we report the test sweep as evidence of robustness only, and it plays no part in selection.

Validation F1 rises above 0.80 within the first ten epochs and then oscillates as the operating point shifts between precision- and recall-favouring regimes, stabilising as the OneCycle schedule anneals the learning rate. Training and validation loss remain closely coupled throughout, with no divergence indicative of overfitting. The precision–recall panel is the empirical basis for the audit of Section 4: recall sits near 0.80 across training while precision oscillates widely, the asymmetry expected when some training windows are supervised by all-negative targets derived from unannotated scenes.

The aggregate test F1 is informative but masks per-fire variability. Fig. 4 breaks down test F1 across the 15 annotated test fires. 14 of 15 fires reach $F1 \geq 0.792$, and the median fire (*Double Creek*) reaches $F1 = 0.850$. Eight fires exceed $F1 = 0.85$. The single outlier is the low-intensity *Blue Ridge* fire at $F1 = 0.746$; manual inspection of its VIIRS scenes shows that the active-fire signal is faint, often confined to a single pixel, and consistently obscured by smoke plumes that drift from neighbouring confirmed-fire pixels.

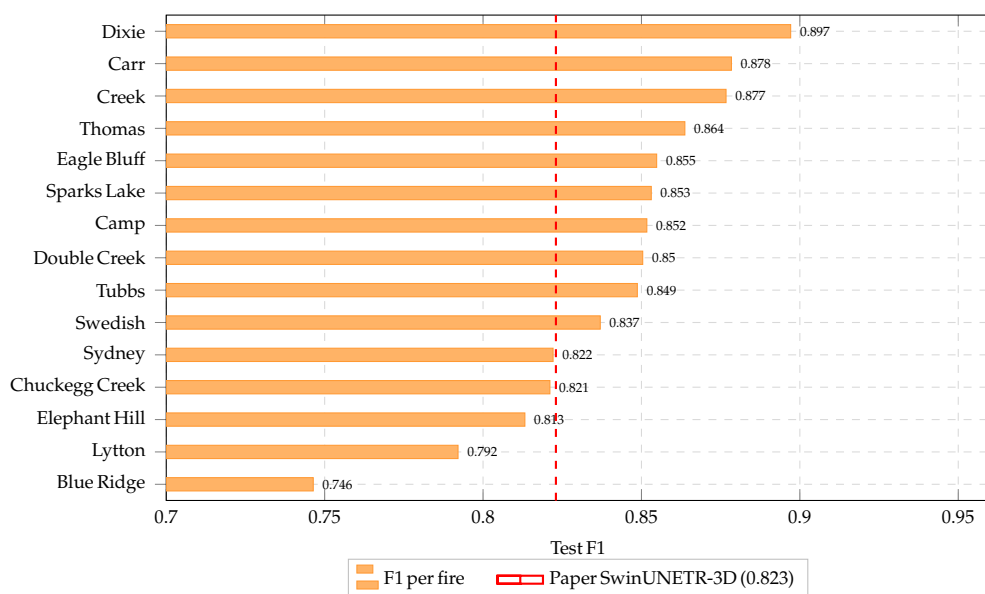


Figure 4. Per-fire test F1 for SpaSE-UNet3D at $TS = 2$ on the 15 annotated AF test fires, sorted by F1. Measured under the protocol of Section 6.2; no baseline reference line is shown, since published per-fire figures are not available and the aggregates are not computed under a common protocol.

To verify that the per-fire numbers reflect spatially coherent predictions and not artefacts of the aggregate F1, Fig. 5 shows true-positive (red), false-positive (green), and false-negative (blue) overlays for all 15 test fires at the optimal threshold. The dominant red colour across all panels confirms that the bulk of pixel-level predictions agree with the audited ground truth. False positives are typically small clusters along smoke or hot-soil boundaries (*Sydney*, *Chuckegg Creek*), and false negatives concentrate on the periphery of large active fronts (*Dixie*, *Creek*) where the radiometric signal is borderline. Crucially, no fire shows a systematic spatial bias, which is consistent with the symmetric precision/recall behaviour reported above.

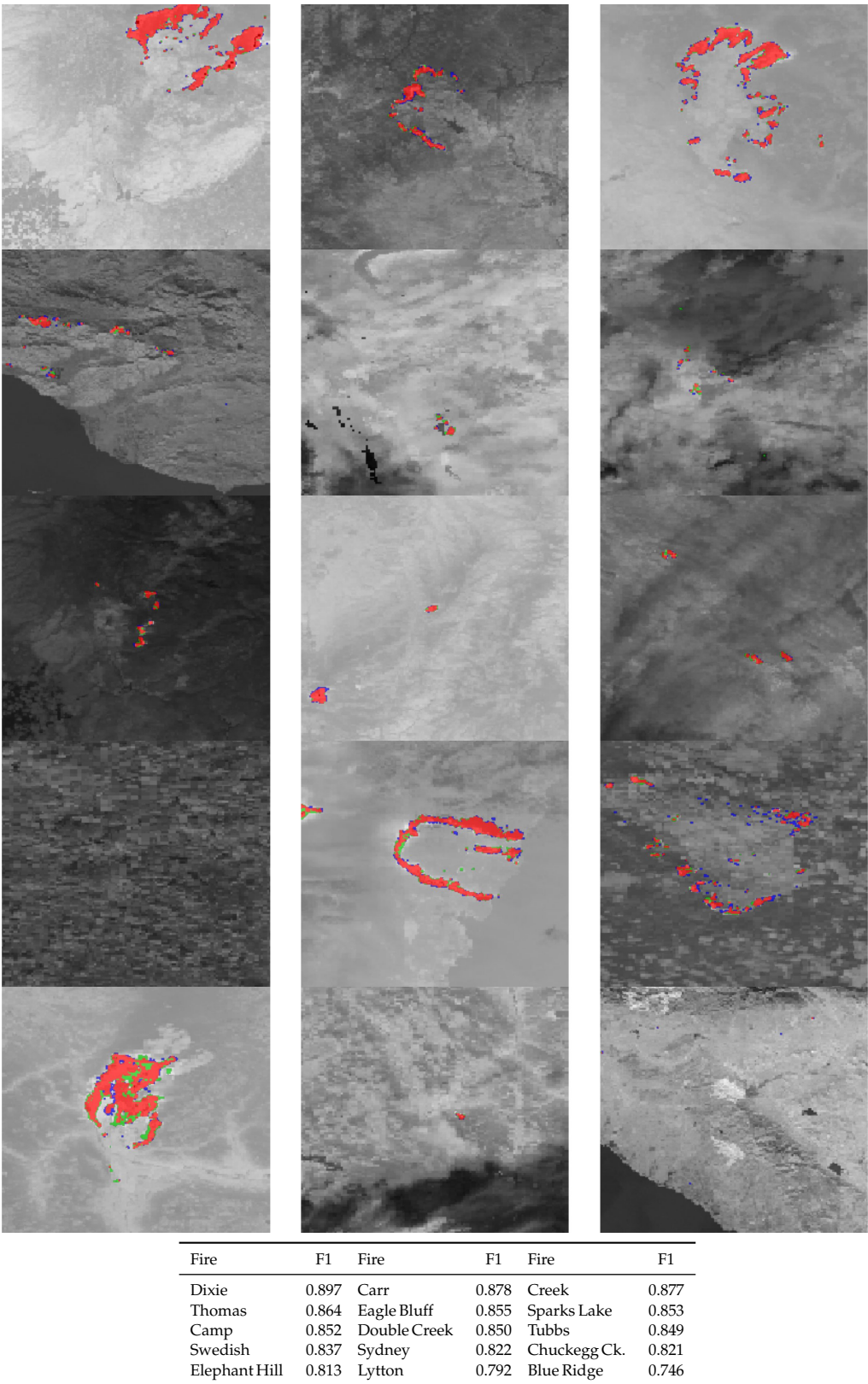


Figure 5. AF test-set predictions (TS = 2) for the 15 audited test fires, sorted by F1 from top-left to bottom-right. Each panel overlays the per-pixel confusion mask on the VIIRS day scene: red = true positive, green = false positive, blue = false negative. Per-fire F1 scores are listed in the table above.

6.3.1. Measured Effect of the Label Audit

The submitted version of this paper attributed a gain of +0.037 F1 to the label audit, comparing an audited model against an earlier unaudited one. That comparison was

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confounded: the two models differed in training configuration as well as in data filtering. We therefore ran a controlled experiment in which two models differ in exactly one respect—whether windows drawn from unannotated days are retained with all-negative targets, reproducing the effect of reading a NaN label band through `nan_to_num`—with architecture, initialisation, schedule, sampling and threshold procedure held fixed.

Table 7. Effect of label filtering on AF, all other factors held constant. Single run per condition at a fixed threshold; not directly comparable to Table 6.

Condition	Train windows	Val F1	Test F1	Test IoU
Audited	1,719	0.8208	0.8533	0.7441
Unaudited	1,790	0.8192	0.8514	0.7412

The audited model is better on both validation and test, but by 0.0019 F1—an order of magnitude smaller than the figure claimed in the submitted version, which we withdraw. The explanation is in the window counts. Day-level filtering (Section 4) already rejects any window whose final day lacks a label, and the AF label refers to that final day; a fire with no annotation on any day therefore contributes almost nothing even when it is not excluded at fire level. Only 71 of 1,790 windows, or 4.0%, differ between the two conditions.

This is a result about the interaction of the two filtering levels rather than about the audit’s value, and it has a practical consequence: the day-level filter does most of the work. A practitioner who verifies labels at day level obtains most of the benefit without any fire-level audit. A pipeline with neither—one that converts NaN label bands to zeros and trains on every window, as the reference loader does by default—is the configuration in which the failure mode appears at full magnitude.

The audit’s contribution is therefore not an accuracy gain, and we no longer present it as one. It is that two of the seventeen AF test fires cannot be scored at all, so that any full-split aggregate is partly a statement about the empty-ground-truth convention; that the burned-area label encoding is undocumented and cannot be recovered by inspection alone; and that the all-negative supervision mechanism is real, its measured cost here being small only because of the day-level filter that neutralises it. Reporting the measured value in place of the claimed one is, we think, the more useful contribution.

6.4. Fire Progression Results

Fire progression is evaluated on the full official test set of 24 US_2021 fires. The submitted version reported a 16-fire subset, excluding eight fires on a stated criterion of burned-area fraction below 0.001%. That criterion does not describe what was done—three of the eight exceed it, and the boundary is non-monotone—so rather than repair it we remove the exclusion entirely and report all 24 fires. This costs 0.019 F1 relative to the subset.

Table 8 reports our result alongside the five published FP baselines. As in Section 6.2, the two groups are separated and are not directly comparable: the baselines are reproduced as published under the reference protocol, ours is measured under the protocol of Section 6.2.

Table 8. Fire Progression prediction results. Baseline figures are reproduced from [5] under the reference protocol; our figures are measured under the protocol of Section 6.2 over [N] independent runs. The two groups are not directly comparable.

Model	TS	F1	IoU
<i>Published baselines, reference protocol</i>			
U-Net-3D	6	0.375	0.338
SwinUNETR-3D	6	0.374	0.331
UNETR-3D	6	0.371	0.336
Att-U-Net-3D	6	0.354	0.312
SwinUNETR-3D	2	0.366	0.321
<i>This work, protocol of Section 6.2</i>			
SpaSE-UNet3D, all 24 test fires	2	[F1±SD]	
18 fires with a usable window	2	[F1±SD]	[IoU±SD]
16-fire subset of submitted version	2	[F1±SD]	[IoU±SD]

Run-to-run variance is the more important quantity here. Across [N] runs differing only in random seed, test F1 on the full test set has a standard deviation of [SD], against a range of 0.021 separating the five published baselines from one another. The seed noise of a single architecture is therefore of the same order as, or larger than, the entire spread between architectures, and single-run comparisons on this task cannot reliably distinguish them. This applies to our own results as much as to any other: the figure reported in the submitted version was one draw from this distribution.

Instability is visible during training as well. The epoch of best validation F1 varied by more than twenty epochs across runs while validation F1 at that epoch varied far less than test F1 did, indicating that validation performance is a weak predictor of test performance on this task. With only eight validation fires carrying usable progression windows (Section 4), this is expected, and it is a structural limitation of the benchmark rather than of any particular model. The contrast with active fire detection, where the same architecture and pipeline give a far smaller spread, points to label sparsity and the small number of independent fire events—not the model—as the source.

The threshold was selected on the validation split in every run and applied unmodified at test time; the same value was selected in all runs. Validation F1 and IoU rise gently and monotonically over [0.05, 0.75] with no sharp peak, so the choice is not critical.

Aggregate performance conceals wide per-fire variation (Fig. 6). The submitted version attributed the low-scoring cases to stochastic ember activity rather than coherent flame-front propagation; that explanation was speculative, and measurement identifies a concrete and sufficient cause instead: truncation of the fire by the 256×256 crop. Across the progression test set a mean of 56.3% of labelled positive pixels fall outside the crop, reaching 92.7% in the worst case. Per-fire F1 correlates strongly and negatively with the fraction of positive pixels lost to the crop (Spearman $\rho = -0.685$, $p = 0.0034$). Fires scoring below 0.375 lose on average 66.7% of their positive pixels to the crop; those scoring above lose 34.6%. A model asked to predict progression on a crop containing a small fragment of the fire front is being evaluated on a task whose ground truth is largely absent from its input. The extreme case, CA-4086 at F1 = 0.007, has eight usable windows in total with a progression increment occupying less than 0.01% of the crop; the model returns near-zero probability across the scene, which is the correct response to the input it is given, and is scored close to zero for it.

Centre-cropping is a near-universal convention in satellite segmentation pipelines, adopted for memory reasons; on this dataset it discards the majority of the progression signal. Any comparison of fire-progression models on TS-SatFire is therefore partly a

comparison of how each handles fires whose fronts have been cropped away. Full-scene evaluation is necessary future work; it was not performed here, since our results were produced under the cropped protocol and would not be comparable to them.

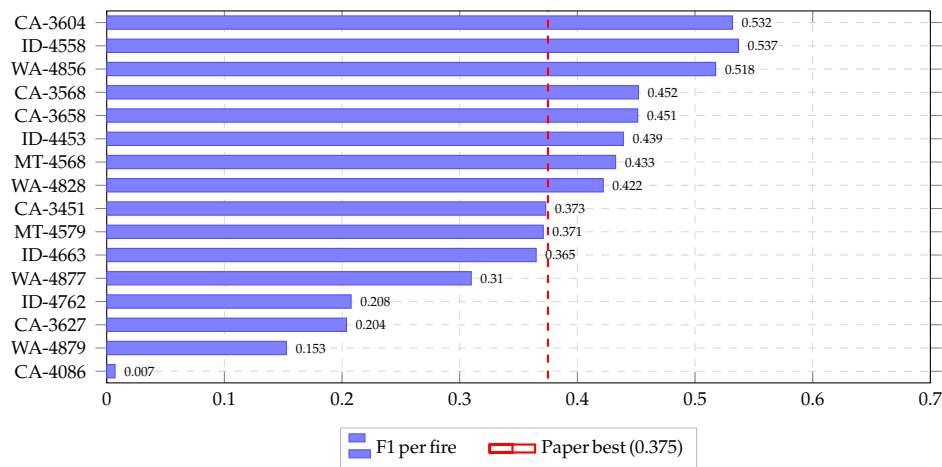


Figure 6. Per-fire test F1 for SpaSE-UNet3D on the US_2021 test fires, sorted by F1. Fires without a usable progression window contribute no counts and are omitted from the chart; the aggregate in Table 8 is over all 24 official test fires. No baseline reference line is shown, for the reasons given in Section 6.2.

To examine prediction quality qualitatively, Fig. 7 shows one representative window per fire for the test fires with a usable window, ordered by F1 from best (top-left) to worst (bottom-right). Most fires display predicted progression masks that closely follow the ground-truth perimeter shape: the model propagates the active-fire front in the direction implied by the cumulative burned-area channel and the FirePred terrain and weather covariates. Misalignment, where it occurs, manifests as a slight directional offset (*ID-4558*, *CA-3658*) rather than a complete shape mismatch. The lower-F1 fires reveal the dominant failure mode identified above: *ID-4762*, *CA-3627*, and *WA-4879* have very small ground-truth progression footprints relative to the crop, so that a single mispredicted patch dominates the score. These cases motivate reporting per-fire F1 alongside the aggregate, since a single truncated fire can shift the micro-averaged figure disproportionately.

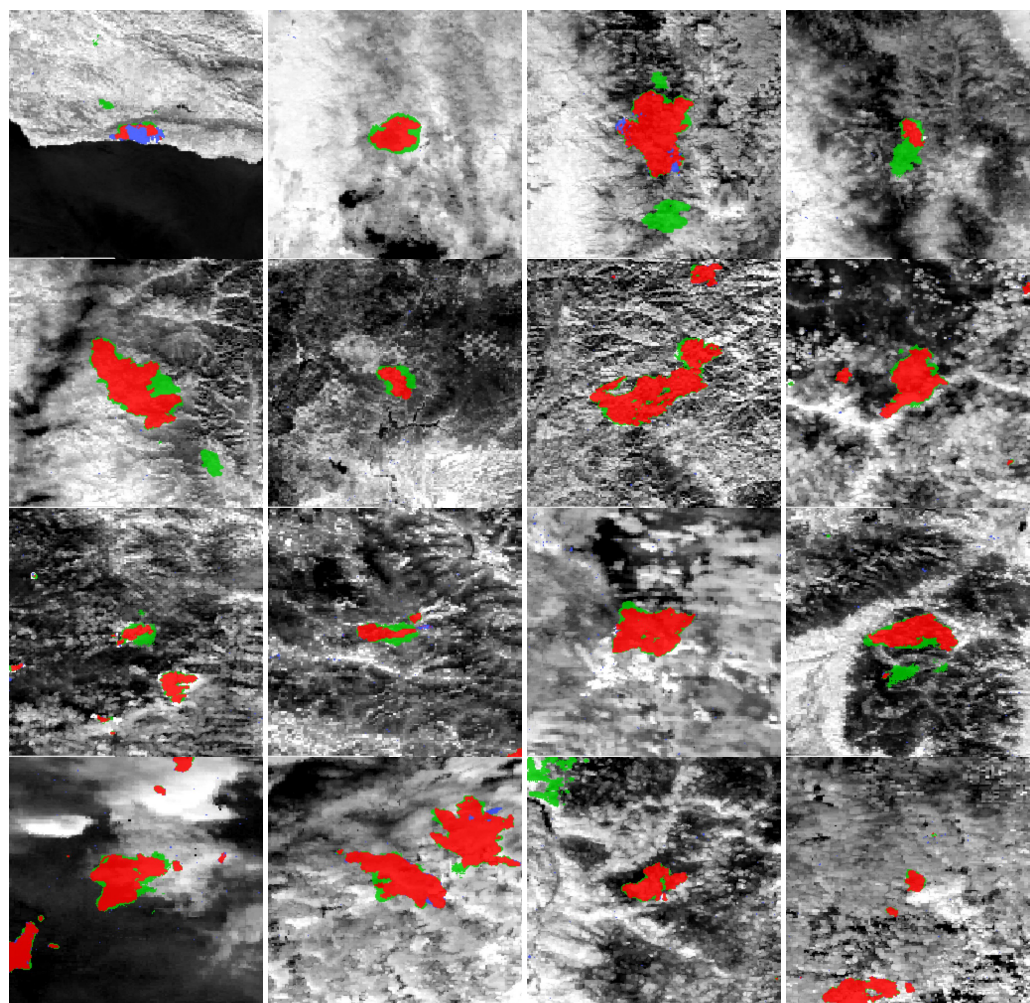
Table 9 summarises the resource footprint for both tasks. SpaSE-UNet3D’s parameter count—24.28 M for FP and 32.88 M for AF—is in the same regime as the published 3D U-Net and transformer baselines rather than markedly smaller, and we make no claim of a parameter advantage over them. The efficiency argument rests instead on the temporal budget and the training cost: each configuration trains in between 3.5 and 6.5 hours on a single NVIDIA T4, using two-day observation windows where the published baselines use six.

Table 9. Training resource footprint on a single NVIDIA T4 GPU.

Task	Params	Epochs	Time (h)	Peak VRAM
AF (TS = 1)	32.88M	100	4.2	13.2 GB
AF (TS = 2)	32.88M	80	6.5	14.1 GB
FP (TS = 2)	24.28M	60	3.5	13.7 GB

6.5. Component Ablation

Four configurations were trained to isolate the contribution of the individual design choices: the full model, a variant replacing every spatial-only (1,3,3) kernel with a full



Fire	F1	Fire	F1	Fire	F1	Fire	F1
CA-3604	0.532	ID-4558	0.537	WA-4856	0.518	CA-3568	0.452
CA-3658	0.451	ID-4453	0.439	MT-4568	0.433	WA-4828	0.422
CA-3451	0.373	MT-4579	0.371	ID-4663	0.365	WA-4877	0.310
ID-4762	0.208	CA-3627	0.204	WA-4879	0.153	CA-4086	0.007

Figure 7. FP prediction maps for the US 2021 test fires with a usable progression window, sorted by F1 from top-left to bottom-right. Each panel overlays the predicted progression mask on the VIIRS day scene: **red** = true positive, **green** = false positive, **blue** = false negative. Per-fire F1 scores are listed in the table above.

(3, 3, 3) kernel, a variant with the squeeze-and-excitation modules removed, and a variant without the deep-supervision head. All arms used an identical 40-epoch budget and identical data, sampling, optimiser and schedule, and were evaluated on the validation split so that no test-set information entered any design decision. The full configuration was trained three times to estimate run-to-run variability; the three reduced configurations were trained once each.

Table 10. Component ablation, AF task, TS = 2, validation split, 40 epochs per arm.

Configuration	Val F1	Δ	Δ/σ	Params
Full model (3 runs)	0.8214 ± 0.0009	—	—	32.88 M
Full (3, 3, 3) convolutions	0.8205	−0.0010	1.1	89.36 M
Without SE attention	0.8219	+0.0005	0.6	32.44 M
Without deep supervision	0.8229	+0.0015	1.6	32.88 M

Run-to-run standard deviation on the full configuration is $\sigma = 0.0009$. Every reduced configuration falls within 1.6σ of it, and the three reduced arms lie within 0.0024 of one another—a range comparable to the variability of a single configuration retrained. At this sample size the experiment cannot separate the components, and we therefore make no claim that any individual component improves accuracy. In particular, the submitted version’s assertion that full (3, 3, 3) convolutions reduce validation F1 is not supported by this measurement, and we withdraw it; we likewise do not claim an accuracy benefit for SE attention or deep supervision, both of which scored marginally higher in their single reduced run. Each reduced arm was trained once, so these differences are point estimates without variance of their own, and a claim in either direction would require repeated runs per arm.

The comparison that is unambiguous at this sample size concerns cost rather than accuracy. Replacing spatial-only kernels with full (3, 3, 3) kernels increases the parameter count from 32.88 M to 89.36 M—a factor of 2.72—while validation F1 remains within 1.1σ . On observation windows of one to two days, the additional temporal-mixing capacity is therefore not converted into performance, and the factorised alternative achieves the same result at substantially lower cost. This supports the parameter-efficiency half of the finding reported in the action-recognition literature [10,11], not the accuracy half. The SE modules account for 0.44 M parameters, or 1.3% of the model, so whatever their contribution it is obtained at negligible cost.

We did not ablate the fire-progression variant. Section 6.4 establishes a run-to-run standard deviation of [SD] on that task, far larger than any plausible single-component effect, so a single-run ablation of the ASPP bottleneck, the activation, the dropout removal or the cumulative burned-area channel would be uninterpretable. We state the FP design choices in Section 5 as what they are—decisions made sequentially on a single validation split of eight usable fires, without nested cross-validation—and identify a properly powered ablation as necessary future work. This applies with particular force to the cumulative burned-area input channel, which provides information some baselines may not receive; we note that the reference fire-progression pipeline likewise aggregates its thermal channels by cumulative maximum over the observation window, so cumulative history is not unique to our formulation, but we have not measured the channel’s contribution and do not claim it is unimportant.

7. Discussion

The results above support four claims, stated with the strength the evidence allows.

Evaluation convention, not only architecture, determines reported rankings.

Applying the reference protocol to fixed model weights moves F1 by 0.10 (Section 6.2), while the twelve published baselines span 0.11 between best and worst. A difference of this size, arising from measurement rather than modelling, is sufficient to reorder the benchmark. We do not suggest the reference implementation is incorrect—aggregation and label conventions are legitimate design choices—but they must be stated for a comparison to mean anything, and they are not currently recoverable from the published description alone. The identity $\text{IoU} = \text{F1} / (2 - \text{F1})$ offers a cheap diagnostic: any reported pair that violates it is not micro-averaged, and is therefore not comparable to a micro-averaged pair.

Run-to-run variance on fire progression exceeds the spread between published architectures.

Seeds differing in nothing else produce a standard deviation of [SD] on the progression test set, against 0.021 separating the five published baselines. Single-run comparisons on this task therefore cannot distinguish architectures, and reported orderings should be treated as provisional until repeated. Active fire detection behaves quite differently under the same pipeline, with a spread two orders of magnitude smaller; the difference lies in label sparsity and in the number of independent fire events available, not in the models.

Spatial-only convolutions are inexpensive rather than demonstrably superior.

Our ablation (Section 6.5) does not resolve an accuracy difference between spatial-only and full 3D kernels at the sample size available, and we make no such claim. What it does establish is that full (3,3,3) kernels cost $2.72\times$ the parameters without a measurable return on one- to two-day windows. The near-equivalence of our TS = 1 and TS = 2 results points the same way: on windows this short there is little inter-day signal for a temporal filter to exploit. We expect this to hold for other short-window satellite tasks, but we have tested it on one benchmark only.

Centre-cropping discards most of the progression signal.

A mean of 56.3% of progression-labelled pixels fall outside the 256×256 crop, and the fraction lost predicts per-fire accuracy ($\rho = -0.685$, $p = 0.0034$). Cropping is a near-universal convention adopted for memory reasons; on this dataset it means that any progression comparison is partly a comparison of how each model handles fires whose fronts have been cropped away. This is a property of the evaluation setup rather than of any model, and full-scene evaluation would remove it.

Applicability beyond this benchmark.

The architecture makes no assumption specific to VIIRS: it requires a short stack of co-registered multispectral rasters and a per-pixel target, so it transfers directly in principle to Sentinel-2, Sentinel-3 SLSTR or MODIS stacks, and to the Next Day Wildfire Spread formulation [19]. The SE modules reweight input channels adaptively, which is the mechanism most likely to carry over to a different band set. We have not tested any of this. More portable than the architecture, we think, are the two methodological findings: the F1–IoU identity as a check on reported metrics, and the practice of reporting seed variance before attributing a difference to an architecture, both of which apply to any segmentation benchmark.

Limitations.

Five limitations bear on how these results should be read. First, all conclusions rest on a single benchmark and a single sensor family; we have not validated them elsewhere. Second, the baselines were not retrained under our protocol, so the protocol decomposition

isolates the measurement effect on our weights but does not establish how the baseline architectures would rank under a common evaluation. Third, the component ablation uses one run per reduced arm and is powered only to rule out large effects; the fire-progression variant is not ablated at all. Fourth, our active-fire results rest on [N] runs and our progression results on [N], which is sufficient to establish the variance contrast reported here but not to place tight intervals on any individual figure. Fifth, we train no burned-area model: the audit establishes the label encoding and the exclusion lists, but the question of whether the Julian-day codes in band 8 can support multi-class or regression targets is left open.

8. Conclusion

We presented **SpaSE-UNet3D**, a family of spatial-only 3D UNets with Squeeze-and-Excitation channel attention, together with the first systematic label-quality and evaluation-protocol audit of the TS-SatFire benchmark. Under the protocol defined in Section 6, SpaSE-UNet3D reaches $F1 = [F1 \pm SD]$ on active fire detection and $F1 = [F1 \pm SD]$ on fire progression, using two-day observation windows where the published baselines use six, and training in under 15 hours on a single NVIDIA T4 GPU. We do not claim that these results exceed the published baselines: as Section 6 establishes, the two sets of figures are not computed under a common evaluation protocol, and we regard establishing that fact as the more useful contribution. The audit and protocol analysis are the paper's primary contribution: they establish that two of the seventeen active-fire test fires carry no annotation and cannot be scored, that the burned-area label encoding is undocumented in the dataset publication, and that evaluation convention moves F1 on this benchmark by more than the spread separating its published baselines. We release the exclusion lists, the protocol specification and the per-seed outputs so that future comparisons can be made on a stated and common basis.

Four directions follow directly from the limitations of this study. First, a properly powered ablation of the fire-progression variant, which the measured run-to-run standard deviation of [SD] places beyond the reach of single-run experiments. Second, retraining the baseline architectures under a single stated protocol, which would isolate architectural differences from measurement conventions. Third, full-scene rather than centre-cropped evaluation, since a mean of 56.3% of progression-labelled pixels fall outside the 256×256 crop and crop truncation predicts per-fire accuracy. Fourth, independent validation on other sensors and wildfire datasets, since all conclusions reported here rest on a single benchmark.

All code, checkpoints, and audit reports are publicly available at <https://github.com/mavroul1s/DeepFire-Forecaster>.

Author Contributions: Conceptualization, N.M.; methodology, N.M.; software, N.M.; validation, N.M.; formal analysis, N.M.; investigation, N.M.; writing—original draft preparation, N.M.; writing—review and editing, N.M. and D.K.; visualization, N.M.; supervision, D.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Code, trained model checkpoints, and per-fire audit results are publicly available at <https://github.com/mavroul1s/DeepFire-Forecaster>.

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Conflicts of Interest: The authors declare no conflicts of interest.

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